BlockerAnalyzer — Per-Shape Hypothesis-Cluster Triage

Source: `bench/dataset.jsonl` (621 rows = branch × shape axes over 554 distinct branches; 257 validated / 364 inconclusive rows). **#br** = validated branches the cluster owns. `*` = unbuilt tool. `[df]` = `decidable:false`.

Shape	Cluster	Cluster description	#br	Tool(s)	Metric(s)	Metric meaning	Verify rule
aflfast_rarity_LW	aflfast_nfuzz_damping_starves_deep_path `[df]`	AFLFast FAST n_fuzz rarity damping de-prioritizes a frequently-hit (saturated) seed and starves the deep path of sustained energy; minimizer's calibration-only schedule (PowerSchedule None) keeps allocating mutations and reaches it	0	energy_share*	fast_energy_share_on_common_edge_seed, minimizer_energy_share_on_common_edge_seed, fast_depth_reach, minimizer_depth_reach	**fast_energy_share_on_common_edge_seed**: fraction of havoc energy/execs the aflfast(fast) schedule spends on the rare-edge seed **minimizer_energy_share_on_common_edge_seed**: fraction of havoc energy/execs the minimizer schedule spends on the rare-edge seed **fast_depth_reach**: max coverage-edge depth past the rare edge reached by aflfast(fast) descendants **minimizer_depth_reach**: max coverage-edge depth past the rare edge reached by minimizer descendants	DECIDABLE:FALSE – no rule can fire. fast_energy_share_on_common_edge_seed and *_depth_reach are NOT obtainable: this is a runtime scheduling dynamic (campaign-wide n_fuzz history + per-seed energy...
aflfast_rarity_LW	*(inconclusive)*	*branches with no validated label*	2	—	—	—	—
aflfast_rarity_WL	rare_edge_energy_overdrive `[df]`	fast over-invests havoc energy in the rare-edge seed, so its corpus shows more exploration past the rare edge than minimizer's	0	energy_share*	rare_seed_descendant_share, rare_seed_depth_reach	**rare_seed_descendant_share**: fraction of the aflfast(fast) corpus that descends from the rare-edge seed **rare_seed_depth_reach**: max coverage-edge depth past the rare edge reached by aflfast(fast) descendants	rare_seed_descendant_share $\geq$ 1.5 AND rare_seed_depth_reach $>$ 0
aflfast_rarity_WL	*(inconclusive)*	*branches with no validated label*	2	—	—	—	—
calibrated_energy_WL	calibrated_structural_assembly	Calibrated energy concentrates havoc on compact/fast/stable seeds, so its corpus assembles a LARGER / more-structured (more-token) input than naive's flat budget ever funds	0	joint_necessity	size_lift, tag_lift	**size_lift**: winner/loser median seed-size ratio (assembly extent the gradient grows) **tag_lift**: winner_tags − loser_tags (structural-token advantage)	size_lift $\geq$ 1.3 AND tag_lift $\geq$ 1.3
calibrated_energy_WL	calibrated_rare_value_targeting	Calibrated energy raises per-seed havoc count on near-valid seeds, raising the probability of landing a specific RARE byte / multi-byte codepoint value at the gating position; the win is value-precision, NOT a larger structure	0	value_distance_reached	value_dist_lift, calibrated_hits_value	**value_dist_lift**: winner/loser ratio of value-distance closed **calibrated_hits_value**: whether the calibrated-energy arm hits the rare target value at all	calibrated_hits_value == true AND value_dist_lift $>$ 1.0
calibrated_energy_WL	*(inconclusive)*	*branches with no validated label*	13	—	—	—	—
ctx_coverage_LW	ctx_corpus_inflation	ctx inflates the corpus / map so naive_ctx hoards far more (context-redundant) entries than naive	0	corpus_size_ratio	corpus_count_ratio	**corpus_count_ratio**: saved-corpus size (or I2S-arm/baseline size ratio) — corpus bloat	corpus_count_ratio $\geq$ 1.5
ctx_coverage_LW	ctx_depth_inflation_at_gate	a high-call-count / multi-context path at the gate is what ctx fragments (flooding) vs a shallow exact-literal gate (dilution)	**2**	depth_reach	call_count_lift	**call_count_lift**: naive_ctx/naive ratio of calls into the gate function (loop flooding vs dilution)	call_count_lift $\geq$ 3.0 (flooding sub-mechanism) vs call_count_lift $<$ 3.0 (dilution sub-mechanism)
ctx_coverage_LW	*(inconclusive)*	*branches with no validated label*	4	—	—	—	—
ctx_coverage_WL	iteration_depth_threshold	naive_ctx accumulates higher loop/recursion iteration depth at the blocker than naive, clearing a count/nesting threshold	**14**	depth_reach	iter_depth_lift, naive_ctx_max_iter, naive_max_iter	**iter_depth_lift**: naive_ctx / naive ratio of max loop/recursion iteration count at the gate **naive_ctx_max_iter**: max loop/recursion iteration count at the gate in the naive_ctx arm **naive_max_iter**: max loop/recursion iteration count at the gate in the naive arm	iter_depth_lift $\geq$ 1.5 AND naive_ctx_max_iter $>$ naive_max_iter
ctx_coverage_WL	context_reach_advantage	naive_ctx's resolving corpus reaches the blocker through call-contexts / path-depths that naive's blocking corpus never reaches	0	depth_reach	ctx_reach_lift, naive_ctx_distinct_contexts, naive_distinct_contexts	**ctx_reach_lift**: winner/loser ratio of seeds (or contexts) reaching the blocker **naive_ctx_distinct_contexts**: # distinct call-contexts the naive_ctx corpus reaches the blocker from **naive_distinct_contexts**: # distinct call-contexts the naive corpus reaches the blocker from	ctx_reach_lift $\geq$ 1.5 AND naive_ctx_distinct_contexts $>$ naive_distinct_contexts
ctx_coverage_WL	*(inconclusive)*	*branches with no validated label*	38	—	—	—	—
grimoire_structural_LW	gap_erases_exact_literal	Grimoire's GeneralizationStage replaces the concrete literal byte (I2S/havoc planted) with a $<$GAP$>$ token; structural recombination never restores it, so a byte-equality gate that literal-preserving naive can satisfy stays blocked under grimoire.	0	token_count	naive_literal_count, grimoire_literal_count, literal_presence_ratio	**naive_literal_count**: # of the exact gate literal occurrences in naive seeds **grimoire_literal_count**: # of the exact gate literal occurrences in grimoire seeds **literal_presence_ratio**: fraction of winner seeds still carrying the exact literal after the mutator	naive_literal_count $\geq$ 1 AND literal_presence_ratio $<$ 0.34
grimoire_structural_LW	fragmentation_starves_depth	Grimoire's re-run tax + structural-recombination produce short, fragmented inputs that never advance the parser far enough to pass a quantitative buffer/depth threshold; naive's longer coherent content satisfies the buffer-ahead condition.	0	joint_necessity	naive_median_size, grimoire_median_size, size_lift	**naive_median_size**: naive median seed length **grimoire_median_size**: grimoire median seed length **size_lift**: winner/loser median seed-size ratio (assembly extent the gradient grows)	size_lift $\geq$ 3.0 AND grimoire_median_size $<$ naive_median_size
grimoire_structural_LW	*(inconclusive)*	*branches with no validated label*	3	—	—	—	—
grimoire_structural_WL	structural_token_assembly	grimoire's grammar mutators place structural tokens/constructs into the corpus that naive's byte havoc cannot, building larger/deeper inputs	**9**	joint_necessity	naive_tags, grimoire_tags, tag_lift, size_lift	**naive_tags**: mean # structural format-tokens (sfnt/OT tags, PNG chunks, SQL kw...) present per seed in the naive arm **grimoire_tags**: mean # structural format-tokens (sfnt/OT tags, PNG chunks, SQL kw...) present per seed in the grimoire arm **tag_lift**: winner_tags − loser_tags (structural-token advantage) **size_lift**: winner/loser median seed-size ratio (assembly extent the gradient grows)	tag_lift $\geq$ 1.0 AND naive_tags $<$ grimoire_tags AND grimoire_tags $\geq$ 2
grimoire_structural_WL	structural_depth_only	fallback sub-hypothesis: grimoire reaches the gate via larger/deeper structure without a target-vocabulary token signal (size lift but flat token count)	**2**	joint_necessity	size_lift	**size_lift**: winner/loser median seed-size ratio (assembly extent the gradient grows)	size_lift $\geq$ 1.3
grimoire_structural_WL	*(inconclusive)*	*branches with no validated label*	31	—	—	—	—

continued on next page

Shape	Cluster	Cluster description	#br	Tool(s)	Metric(s)	Metric meaning	Verify rule
i2s_vp_LLWL	joint_assembly_depth	vpc assembles a LARGER, multi-field structure than I2S-only (cmplog) ever reaches; the gradient is necessary to grow size/token-count past where exact substitution stalls, and I2S is necessary because the gradient alone (value_profile) places $\sim$0 structural tokens	**6**	joint_necessity	vp_tags, size_lift	**vp_tags**: mean # structural format-tokens (sfnt/OT tags, PNG chunks, SQL kw...) present per seed in the value_profile arm **size_lift**: winner/loser median seed-size ratio (assembly extent the gradient grows)	vp_tags $<$ 0.3 AND size_lift $\geq$ 1.3
i2s_vp_LLWL	joint_value_precision	vpc and cmplog reach a SIMILARLY-sized structure (size_lift $\sim$ 1, sometimes vpc smaller), but vpc places a specific gradient-climbed VALUE that the cmplog-blocking seeds approach yet never reach; I2S is still necessary to place the surrounding exact literal(s), and the gradient is necessary for the last-mile value precision the structural proxy is blind to	**6**	value_distance_reached	vpc_min_distance, cmp_min_distance, distance_closure_ratio	**vpc_min_distance**: min Hamming/value distance from value_profile_cmplog seeds to the gate literal (0 = gate solved) **cmp_min_distance**: min Hamming/value distance from cmplog/I2S seeds to the gate literal (0 = gate solved) **distance_closure_ratio**: fraction of the value gap the winner closes (vs the loser/naive residual)	size_lift $<$ 1.3 AND cmp_min_distance $>$ 0 AND distance_closure_ratio $\geq$ 0.5
i2s_vp_LLWL	*(inconclusive)*	*branches with no validated label*	11	—	—	—	—
i2s_vp_LLW_	joint_assembly_depth	vpc assembles a LARGER, multi-field structure than I2S-only (cmplog) ever reaches; the gradient is necessary to grow size/token-count past where exact substitution stalls, and I2S is necessary because the gradient alone (value_profile) places $\sim$0 structural tokens	**1**	joint_necessity	vp_tags, size_lift	**vp_tags**: mean # structural format-tokens (sfnt/OT tags, PNG chunks, SQL kw...) present per seed in the value_profile arm **size_lift**: winner/loser median seed-size ratio (assembly extent the gradient grows)	vp_tags $<$ 0.3 AND size_lift $\geq$ 1.3
i2s_vp_LLW_	joint_value_precision	vpc and cmplog reach a SIMILARLY-sized input (size_lift $\sim$ 1), but vpc lands a specific gradient-climbed VALUE (a single exact multi-byte keyword/token) that the cmplog-blocking seeds approach byte-by-byte yet never reach; I2S is still necessary to splice the keyword operand, and the gradient is necessary for the last-mile prefix precision that the structural size proxy is blind to	**9**	value_distance_reached	vpc_min_distance, cmp_min_distance, distance_closure_ratio	**vpc_min_distance**: min Hamming/value distance from value_profile_cmplog seeds to the gate literal (0 = gate solved) **cmp_min_distance**: min Hamming/value distance from cmplog/I2S seeds to the gate literal (0 = gate solved) **distance_closure_ratio**: fraction of the value gap the winner closes (vs the loser/naive residual)	size_lift $<$ 1.3 AND cmp_min_distance $>$ 0 AND distance_closure_ratio $\geq$ 0.5
i2s_vp_LLW_	*(inconclusive)*	*branches with no validated label*	6	—	—	—	—
i2s_vp_LWLL	vp_gradient_distance_climb	value_profile's CMP_MAP gradient climbs a comparison-operand distance ramp toward a structural/syntactic gate state that cmplog- and naive-blocking seeds never approach	**1**	value_distance_reached	vp_min_distance, naive_min_distance, cmplog_min_distance, vp_gate_cmp_satisfied_frac, naive_gate_cmp_satisfied_frac	**vp_min_distance**: min Hamming/value distance from value_profile seeds to the gate literal (0 = gate solved) **naive_min_distance**: min Hamming/value distance from naive seeds to the gate literal (0 = gate solved) **cmplog_min_distance**: min Hamming/value distance from cmplog/I2S seeds to the gate literal (0 = gate solved) **vp_gate_cmp_satisfied_frac**: fraction of value_profile seeds satisfying the gate's CMP at the offset **naive_gate_cmp_satisfied_frac**: fraction of naive seeds satisfying the gate's CMP at the offset	vp_min_distance $<$ cmplog_min_distance AND vp_min_distance $<$ naive_min_distance AND vp_gate_cmp_satisfied_frac $\geq$ naive_gate_cmp_satisfied_frac + 0.2
i2s_vp_LWLL	*(inconclusive)*	*branches with no validated label*	4	—	—	—	—
i2s_vp_LWLW	i2s_decoy_substitution_overfit	I2S substitutes a fixed-offset structural literal (decoy) that pins the loser corpus to the valid-structure path and away from the winner's free-byte target	**7**	operand_enrichment	signed_target_enrich, decoy_enrich	**signed_target_enrich**: log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive ($>$0=enriched, $<$0=depleted) **decoy_enrich**: same log2 ratio for the loser/decoy gate-byte ($>$0=I2S corpus pinned to decoys)	signed_target_enrich $<$ -0.3
i2s_vp_LWLW	i2s_homogenizes_corpus_starves_state_diversity	I2S operand substitution collapses corpus composition onto a few CMP-matched byte patterns, inflating/homogenizing the I2S-arm corpus so it cannot build the diverse runtime state (linked-list population) the structural predicate needs	0	corpus_size_ratio	corpus_size_ratio, composition_entropy_ratio	**corpus_size_ratio**: saved-corpus size (or I2S-arm/baseline size ratio) — corpus bloat **composition_entropy_ratio**: byte-value Shannon entropy of the I2S-arm corpus $\backslash div$ baseline ($<$1 = homogenized)	corpus_size_ratio $\geq$ 1.3 AND composition_entropy_ratio $<$ 0.85
i2s_vp_LWLW	i2s_no_operand_budget_drain `[df]`	Gate is a table-/state-machine-derived or range value with NO fixed-offset input-byte CMP operand; I2S substitution finds nothing to copy and its overhead dilutes the havoc budget that would otherwise land the codepoint by random exploration	0	operand_enrichment	signed_target_enrich	**signed_target_enrich**: log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive ($>$0=enriched, $<$0=depleted)	false
i2s_vp_LWLW	i2s_anchors_large_valid_blocks_truncation `[df]`	I2S (and ctx) anchor the corpus to large valid-header inputs; the branch needs an input shorter than a 128-byte minimum that only plain havoc byte-deletion produces	0	corpus_size_ratio	median_seed_length_ratio	**median_seed_length_ratio**: winner/loser median seed length	false
i2s_vp_LWLW	*(inconclusive)*	*branches with no validated label*	6	—	—	—	—
i2s_vp_LWL_	vp_gradient_distance_closure	value_profile's CMP_MAP gradient incrementally closes byte-distance to a fixed multi-byte literal gate that cmplog-only and vpc never reach; the gradient (not I2S substitution) is the necessary technique, and adding I2S in vpc diverts energy so it stalls farther from the operand	0	value_distance_reached	vp_min_distance, cmp_min_distance, vpc_min_distance, distance_closure_ratio	**vp_min_distance**: min Hamming/value distance from value_profile seeds to the gate literal (0 = gate solved) **cmp_min_distance**: min Hamming/value distance from cmplog/I2S seeds to the gate literal (0 = gate solved) **vpc_min_distance**: min Hamming/value distance from value_profile_cmplog seeds to the gate literal (0 = gate solved) **distance_closure_ratio**: fraction of the value gap the winner closes (vs the loser/naive residual)	vp_min_distance $<$ cmp_min_distance AND vp_min_distance $<$ vpc_min_distance AND cmp_min_distance $>$ 0 AND distance_closure_ratio $\geq$ 0.5
i2s_vp_LWL_	*(inconclusive)*	*branches with no validated label*	2	—	—	—	—
i2s_vp_LWWL	vp_gradient_value_distance_closed	value_profile's CMP_MAP gradient drives its seeds measurably closer to the gate's target value than cmplog/naive seeds get, while I2S substitution does not close the distance	**1**	value_distance_reached	vp_min_dist, cmp_min_dist, dist_gap	**vp_min_dist**: min Hamming/value distance from value_profile seeds to the gate literal (0 = gate solved) **cmp_min_dist**: min Hamming/value distance from cmplog/I2S seeds to the gate literal (0 = gate solved) **dist_gap**: winner$-$loser gap in value/distance closed toward the gate	dist_gap $\geq$ 0.25 AND vp_min_dist $\leq$ 0.35 AND cmp_min_dist $\geq$ 0.50

continued on next page

Shape	Cluster	Cluster description	#br	Tool(s)	Metric(s)	Metric meaning	Verify rule
i2s_vp_LWWL	vp_gradient_corpus_reaches_ deeper	value_profile's gradient-built corpus reaches a deeper / later parse state than cmplog/naive, corroborating the climb when value-distance is unavailable or the gate is a multi-stage parse depth rather than a single operand	0	depth_reach	vp_depth, cmp_depth, depth_lift	**vp_depth**: max loop/recursion/path depth reached at the gate by the value_profile arm **cmp_depth**: max loop/recursion/path depth reached at the gate by the cmplog/I2S arm **depth_lift**: max loop/recursion/path depth reached at the gate by the arm	depth_lift $\geq$ 0.20 AND vp_depth $\geq$ 0.60
i2s_vp_LWWL	*(inconclusive)*	*branches with no validated label*	16	—	—	—	—
i2s_vp_LWWW	i2s_anti_structural_byte_corruption	I2S RandReplace overwrites the exact structural/category/literal bytes naive havoc assembles, depleting (not enriching) the target byte in cmplog's corpus relative to naive	**7**	operand_enrichment	signed_target_enrich	**signed_target_enrich**: log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive ($>$0=enriched, $<$0=depleted)	signed_target_enrich $<$ -0.3
i2s_vp_LWWW	i2s_anti_no_input_preimage_ misdirection	Gate has no input-side CMP preimage (category-table / state-machine classification, length threshold, or internally-derived overflow sentinel); I2S substitution finds no fixed-offset value to copy, so it misallocates budget while naive havoc explores freely	0	operand_enrichment	signed_target_enrich	**signed_target_enrich**: log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive ($>$0=enriched, $<$0=depleted)	signed_target_enrich == no_gate_signature
i2s_vp_LWWW	vp_pro_cmpmap_gradient_rescue	VP CMP_MAP Hamming/prefix-distance gradient gives graded partial-progress feedback toward the gate comparand, letting value_profile/vpc retain near-miss inputs that approach the target while cmplog's edge-only binary feedback stalls	**1**	value_distance_ reached	vp_min_value_distance, cmp_min_value_distance, distance_lift	**vp_min_value_distance**: min Hamming/value distance from value_profile seeds to the gate literal (0 = gate solved) **cmp_min_value_distance**: min Hamming/value distance from cmplog/I2S seeds to the gate literal (0 = gate solved) **distance_lift**: winner − loser gap in value/distance closed toward the gate	distance_lift $\geq$ 1.3 AND vp_min_value_ distance $<$ cmp_min_value_distance
i2s_vp_LWWW	*(inconclusive)*	*branches with no validated label*	3	—	—	—	—
i2s_vp_LWW_	vp_gradient_value_distance_climb	value_profile's CMP_MAP Hamming/prefix-distance gradient drives the corpus to smaller residual value-distance at the gating comparison site(s), so vp/vpc reach the comparison target where cmplog (edge-only, no gradient) stalls at a larger distance	0	value_distance_ reached	vp_min_distance, cmp_min_ distance, distance_closure_ ratio	**vp_min_distance**: min Hamming/value distance from value_ profile seeds to the gate literal (0 = gate solved) **cmp_min_distance**: min Hamming/value distance from cmplog/I2S seeds to the gate literal (0 = gate solved) **distance_closure_ratio**: fraction of the value gap the winner closes (vs the loser/naive residual)	distance_closure_ratio $\geq$ 1.5 AND vp_min_ distance $<$ cmp_min_distance
i2s_vp_LWW_	vp_gradient_depth_reach_ corroborant	corroborating coverage signal: value_ profile/vpc reach a deeper point past the gated comparison earlier than cmplog, target-independently (no corpus needed)	0	depth_reach	vp_reach_frac, cmp_reach_ frac, reach_lift	**vp_reach_frac**: fraction of value_profile seeds that reach the gate region **cmp_reach_frac**: fraction of cmplog/I2S seeds that reach the gate region **reach_lift**: winner/loser ratio of seeds (or contexts) reaching the blocker	reach_lift $\geq$ 0.5
i2s_vp_LWW_	*(inconclusive)*	*branches with no validated label*	9	—	—	—	—
i2s_vp_LW_L	vp_gradient_climbs_numeric_ threshold	value_profile-resolving seeds drive a derived NUMERIC field past an inequality/range threshold; there is no exact literal to splice, so distance is the MAGNITUDE GAP to crossing the threshold – vp closes it monotonically while cmplog (no single constant to substitute) and naive (no gradient) stall below the bound	**4**	value_distance_ reached	vp_gap, cmp_gap, naive_ gap, mag_gap_lift, vp_ crossed_frac	**vp_gap**: value_profile residual gap to the numeric threshold value **cmp_gap**: cmplog/I2S residual gap to the numeric threshold value **naive_gap**: naive residual gap to the numeric threshold value **mag_gap_lift**: winner − loser gap in value/distance closed toward the gate **vp_crossed_frac**: fraction of value_profile seeds that cross the numeric threshold	vp_crossed_frac $\geq$ 0.5 AND mag_gap_lift $>$ 0 AND cmp_gap $>$ vp_gap AND naive_gap $>$ vp_gap
i2s_vp_LW_L	vp_gradient_assembles_multibyte_ literal	value_profile-resolving seeds reach a multi-byte operator/keyword/structural LITERAL strictly closer than cmplog-blocking AND naive-blocking seeds; the operand is not a single spliceable constant, so I2S stalls and naive gets no gradient while the CMP_MAP distance bucket ratchets a prefix/Hamming approach to the literal	0	value_distance_ reached	vp_dist, cmp_dist, naive_dist, lit_dist_gap, vp_dist_frac	**vp_dist**: min Hamming/value distance from value_profile seeds to the gate literal (0 = gate solved) **cmp_dist**: min Hamming/value distance from cmplog/I2S seeds to the gate literal (0 = gate solved) **naive_dist**: min Hamming/value distance from naive seeds to the gate literal (0 = gate solved) **lit_dist_gap**: winner − loser gap in value/distance closed toward the gate **vp_dist_frac**: value_profile min distance as a fraction of the full gate width	lit_dist_gap $\geq$ 1.0 AND vp_dist_frac $\leq$ 0.34 AND cmp_dist $>$ vp_dist AND naive_dist $>$ vp_dist
i2s_vp_LW_L	*(inconclusive)*	*branches with no validated label*	3	—	—	—	—
i2s_vp_LW_W	i2s_decoy_substitution_target_ depletion	I2S splices a WRONG/competing but abundant CMP operand into the input (a well-known font-table tag, a script-tag/codepoint constant, an in-range fixed-point magnitude, or a competing protocol value), pinning the cmplog corpus onto that decoy and DEPLETING it of the target byte value at the winner gate's offset; naive/value_profile never harvest the decoy so they freely reach the target	**24**	operand_enrichment	signed_target_enrich, decoy_ enrich	**signed_target_enrich**: log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive ($>$0=enriched, $<$0=depleted) **decoy_enrich**: same log2 ratio for the loser/decoy gate-byte ($>$0=I2S corpus pinned to decoys)	skip != 'no_gate_signature' AND signed_ target_enrich $<$ -0.3
i2s_vp_LW_W	i2s_disrupts_havoc_sequence_ accumulation	The winner gate needs havoc to BUILD or PRESERVE a long / periodically-repeated glyph-record byte sequence (multi-matra record runs, long multi-glyph Thai state-machine traversal); the gate value is table/state-derived (no harvestable operand), but cmplog's CMP-biased budget produces SHORTER, heterogeneous inputs that never reach the required sequence length, while naive's byte-expansion/copy operators grow and sustain it	0	joint_necessity	size_lift	**size_lift**: winner/loser median seed-size ratio (assembly extent the gradient grows)	skip == 'no_gate_signature' AND size_lift $\geq$ 1.3

continued on next page

Shape	Cluster	Cluster description	#br	Tool(s)	Metric(s)	Metric meaning	Verify rule
i2s_vp_LW_W	i2s_overhead_on_derived_gate_no_operand	The gate value is computed from shaping tables, Unicode-property lookups, Ragel state-machine internals, or derived category classification – NOT a literal value compared against input bytes anywhere. I2S has no input-resident CMP operand to harvest or substitute; its per-execution callback overhead and misdirected substitution budget crowd out the broad havoc byte exploration that resolves the branch, with NO directional corpus depletion and NO length asymmetry	0	operand_enrichment	signed_target_enrich	**signed_target_enrich**: log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive (>0=enriched, <0=depleted)	skip == 'no_gate_signature' AND size_lift < 1.3
i2s_vp_LW_W	*(inconclusive)*	*branches with no validated label*	5	—	—	—	—
i2s_vp_L_LW	i2s_case_constant_decoy_substitution	I2S keeps copying enumerated switch case-constants (decoys) into the option-selector byte, pinning control onto a matching case and depleting cmplog/vpc corpora of the non-enumerated values that fall through to default; plain havoc drifts onto default freely	**1**	operand_enrichment	signed_target_enrich, decoy_enrich	**signed_target_enrich**: log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive (>0=enriched, <0=depleted) **decoy_enrich**: same log2 ratio for the loser/decoy gate-byte (>0=I2S corpus pinned to decoys)	signed_target_enrich < -0.3
i2s_vp_L_LW	*(inconclusive)*	*branches with no validated label*	0	—	—	—	—
i2s_vp_L_WL	vp_gradient_value_distance_climb	vpc's CMP_MAP gradient ratchets the corpus byte-by-byte toward a CONCRETE literal token that cmplog's edge-only signal cannot bank near-miss progress toward	**5**	value_distance_reached	w_min_hamming_to_literal, l_min_hamming_to_literal, distance_gap	**w_min_hamming_to_literal**: (derived metric — see hypothesis prediction/params) **l_min_hamming_to_literal**: (derived metric — see hypothesis prediction/params) **distance_gap**: winner−loser gap in value/distance closed toward the gate	distance_gap ≥ 0.25 AND w_min_hamming_to_literal ≤ 0.30
i2s_vp_L_WL	vp_gradient_structural_depth_assembly	vpc's CMP_MAP gradient accumulates partial credit across chained comparisons to drive the corpus to a structurally DEEPER parse state (multi-child content model / namespace binding / internal subset / pointer-identity) whose gate is a derived pointer/state check with NO fixed literal pre-image	0	depth_reach	w_depth, l_depth, depth_lift	**w_depth**: max loop/recursion/path depth reached at the gate by the winner arm **l_depth**: max loop/recursion/path depth reached at the gate by the loser arm **depth_lift**: max loop/recursion/path depth reached at the gate by the arm	depth_lift ≥ 1.5 AND w_depth > l_depth
i2s_vp_L_WL	*(inconclusive)*	*branches with no validated label*	9	—	—	—	—
i2s_vp_L_W_	vp_gradient_climbs_multibyte_cmp_beyond_i2s	vpc-resolving seeds reach a multi-byte operand value strictly closer to the target than cmplog-blocking seeds; the operand is not a single spliceable literal, so I2S stalls while the CMP_MAP gradient ratchets distance down	0	value_distance_reached	vpc_dist, cmp_dist, dist_gap, vpc_dist_frac	**vpc_dist**: min Hamming/value distance from value_profile_cmplog seeds to the gate literal (0 = gate solved) **cmp_dist**: min Hamming/value distance from cmplog/I2S seeds to the gate literal (0 = gate solved) **dist_gap**: winner−loser gap in value/distance closed toward the gate **vpc_dist_frac**: value_profile_cmplog min distance as a fraction of the full gate width	dist_gap ≥ 1.0 AND vpc_dist_frac ≤ 0.34 AND cmp_dist > vpc_dist
i2s_vp_L_W_	*(inconclusive)*	*branches with no validated label*	7	—	—	—	—
i2s_vp_L__W	i2s_energy_diversion_decoy_pin	the gate is an ungradable fixed-offset equality; plain havoc places the literal by random mutation while cmplog diverts energy and leaves its corpus depleted of (pinned away from) the target value at the gate offset	**2**	operand_enrichment	signed_target_enrich, decoy_enrich	**signed_target_enrich**: log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive (>0=enriched, <0=depleted) **decoy_enrich**: same log2 ratio for the loser/decoy gate-byte (>0=I2S corpus pinned to decoys)	signed_target_enrich < -0.3
i2s_vp_L__W	i2s_corpus_channeling_starves_target	cmplog solves upstream structural gates and channels the corpus into a deep valid-decode path, starving the shallow target byte of the random mutation plain havoc supplies incidentally	0	corpus_size_ratio	size_ratio_cmp_over_naive, target_reach_ratio_cmp_over_naive	**size_ratio_cmp_over_naive**: saved-corpus size (or I2S-arm/baseline size ratio) — corpus bloat **target_reach_ratio_cmp_over_naive**: max loop/recursion/path depth reached at the gate by the arm	size_ratio_cmp_over_naive ≥ 1.5 AND target_reach_ratio_cmp_over_naive < 0.5
i2s_vp_L__W	*(inconclusive)*	*branches with no validated label*	0	—	—	—	—
i2s_vp_WLWL	i2s_fixed_offset_literal_value_gate	cmplog (I2S) substitutes a logged CMP literal at a fixed gate offset, so its corpus is enriched in the target value at that offset relative to naive; value_profile/naive cannot synthesize the exact literal	**63**	operand_enrichment	signed_target_enrich, decoy_enrich	**signed_target_enrich**: log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive (>0=enriched, <0=depleted) **decoy_enrich**: same log2 ratio for the loser/decoy gate-byte (>0=I2S corpus pinned to decoys)	signed_target_enrich ≥ 0.8
i2s_vp_WLWL	i2s_multi_literal_scaffold_state_gate `[df]`	I2S assembles two or more co-dependent, dispersed CMP literals (URL scheme+port+HTTP tokens; cookie/header chains; deep multi-field ICC structure) whose accumulated downstream program state, not the substitution point itself, controls the blocked branch	0	value_enrichment*	any_anchor_token_enrich	**any_anchor_token_enrich**: log2 enrichment of the relevant token/literal in the arm vs baseline	any_anchor_token_enrich ≥ 0.8
i2s_vp_WLWL	*(inconclusive)*	*branches with no validated label*	43	—	—	—	—
i2s_vp_WLW_	i2s_literal_operand_substitution	I2SRandReplace splices the exact logged comparand literal into the fixed gate offset; VP's CMP_MAP Hamming gradient cannot reach the exact value within budget	**3**	operand_enrichment	signed_target_enrich, decoy_enrich	**signed_target_enrich**: log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive (>0=enriched, <0=depleted) **decoy_enrich**: same log2 ratio for the loser/decoy gate-byte (>0=I2S corpus pinned to decoys)	signed_target_enrich ≥ 1.0
i2s_vp_WLW_	*(inconclusive)*	*branches with no validated label*	4	—	—	—	—
i2s_vp_WL_L	i2s_exact_literal_constant_gate	cmplog plants an exact fixed-offset literal/magic constant the VP gradient cannot climb to	**3**	operand_enrichment	signed_target_enrich, decoy_enrich	**signed_target_enrich**: log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive (>0=enriched, <0=depleted) **decoy_enrich**: same log2 ratio for the loser/decoy gate-byte (>0=I2S corpus pinned to decoys)	signed_target_enrich ≥ 1.0 AND decoy_enrich ≤ 0.0
i2s_vp_WL_L	i2s_derived_signed_operand_construction	cmplog constructs a DERIVED negative/range value across parser-routing bytes (no stored literal)	0	operand_enrichment	signed_target_enrich	**signed_target_enrich**: log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive (>0=enriched, <0=depleted)	no_gate_signature == true (operand_enrichment returns no fixed-offset literal); hypothesis confirmed by the diagnostic ABSENCE of literal enrichment under the i2s-pro shape, falling to inconclusive...

continued on next page

Shape	Cluster	Cluster description	#br	Tool(s)	Metric(s)	Metric meaning	Verify rule
i2s_vp_WL_L	*(inconclusive)*	*branches with no validated label*	4	—	—	—	—
i2s_vp_WWLW	vpc_corpus_inflation_interference `[df]`	vpc's combined I2S+VP corpus admission inflates and collapses composition, starving the single-technique-found path that both cmplog-alone and value_profile-alone sustain	0	corpus_size_ratio	vpc_corpus_size, cmp_ corpus_size, vp_corpus_ size, vpc_target_value_ fraction	**vpc_corpus_size**: value_profile_cmplog saved-corpus size (or I2S-arm/baseline size ratio) — corpus bloat **cmp_corpus_size**: cmplog/I2S saved-corpus size (or I2S-arm/baseline size ratio) — corpus bloat **vp_corpus_size**: value_profile saved-corpus size (or I2S-arm/baseline size ratio) — corpus bloat **vpc_target_value_fraction**: (derived metric — see hypothesis prediction/params)	vpc_corpus_size / max(cmp_corpus_size, vp_ corpus_size) $\geq$ 1.3 AND vpc_target_value_ fraction $<$ 0.5 * min(cmp_target_value_ fraction, vp_target_value_fraction)
i2s_vp_WWLW	*(inconclusive)*	*branches with no validated label*	2	—	—	—	—
i2s_vp_WWWL	literal_value_gate_cmp_solvable	A fixed-offset literal value gate (FOURCC/tag/enum/single byte) is solved by any CMP-aware fuzzer (I2S substitution or VP Hamming gradient) while naive's blind havoc cannot place the literal	**4**	operand_enrichment	signed_target_enrich, decoy_ enrich	**signed_target_enrich**: log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive ($>$0=enriched, $<$0=depleted) **decoy_enrich**: same log2 ratio for the loser/decoy gate-byte ($>$0=I2S corpus pinned to decoys)	signed_target_enrich $\geq$ 1.0 AND decoy_ enrich $\leq$ 0.0
i2s_vp_WWWL	*(inconclusive)*	*branches with no validated label*	10	—	—	—	—
i2s_vp_WW_L	parallel_fixed_offset_literal_gate	A fixed-offset 1-4 byte VALUE literal/enum gates the branch; both I2S (direct substitution) and VP (Hamming/prefix gradient) drive the same gate byte(s) to the target value, so both resolving corpora are enriched in the target byte at the gate offset relative to naive.	0	operand_enrichment	signed_target_enrich, decoy_ enrich	**signed_target_enrich**: log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive ($>$0=enriched, $<$0=depleted) **decoy_enrich**: same log2 ratio for the loser/decoy gate-byte ($>$0=I2S corpus pinned to decoys)	cmplog.signed_target_enrich $>$ 0.5 AND value_profile.signed_target_enrich $>$ 0.5
i2s_vp_WW_L	parallel_structural_accumulation_ gate	The gate is NOT a single fixed-offset literal but a structural/length/cascade condition (repeated integer-token accumulation, or a multi-comparison chunk-validation cascade). Both I2S and VP help by accumulating/clearing many comparisons, so the resolving seeds are larger / carry more target tokens than the blocking seeds — there is no single alignable gate byte.	0	joint_necessity	size_lift, tag_lift	**size_lift**: winner/loser median seed-size ratio (assembly extent the gradient grows) **tag_lift**: winner_tags $-$ loser_tags (structural-token advantage)	size_lift $\geq$ 1.3 AND tag_lift $\geq$ 1.3
i2s_vp_WW_L	*(inconclusive)*	*branches with no validated label*	9	—	—	—	—
i2s_vp_W_WL	i2s_fixed_offset_literal_substitution	I2S enriches the cmplog corpus with the exact CMP-target byte(s) at the gate offset (literal substitution), which naive's blind havoc cannot place	**18**	operand_enrichment	signed_target_enrich, decoy_ enrich	**signed_target_enrich**: log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive ($>$0=enriched, $<$0=depleted) **decoy_enrich**: same log2 ratio for the loser/decoy gate-byte ($>$0=I2S corpus pinned to decoys)	signed_target_enrich $\geq$ 1.0
i2s_vp_W_WL	*(inconclusive)*	*branches with no validated label*	38	—	—	—	—
i2s_vp_W__L	i2s_literal_substitution_gate	A fixed-offset literal CMP gate that I2S satisfies by splicing the logged operand byte(s) into the input; cmplog's corpus is enriched for the target literal at the gate offset where naive's is not	0	operand_enrichment	signed_target_enrich, decoy_ enrich	**signed_target_enrich**: log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive ($>$0=enriched, $<$0=depleted) **decoy_enrich**: same log2 ratio for the loser/decoy gate-byte ($>$0=I2S corpus pinned to decoys)	gate_signature == true AND signed_target_ enrich $\geq$ 1.0
i2s_vp_W__L	i2s_relational_collision_gate	A relational / structural gate (hash-bucket chain growth) with NO single byte-traceable operand: I2S substitutes name-equality operands to manufacture COLLIDING entries, so the decisive feature is multiplicity of related names, not one literal at one offset	0	operand_enrichment	signed_target_enrich	**signed_target_enrich**: log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive ($>$0=enriched, $<$0=depleted)	gate_signature == false
i2s_vp_W__L	*(inconclusive)*	*branches with no validated label*	7	—	—	—	—
i2s_vp__LWL	i2s_exact_literal_gate	vpc resolves an exact-literal equality gate because its I2SRandReplace splices the logged comparand literal into a fixed input offset, which value_profile's Hamming/CMP_ MAP gradient cannot land	**55**	operand_enrichment	signed_target_enrich, decoy_ enrich	**signed_target_enrich**: log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive ($>$0=enriched, $<$0=depleted) **decoy_enrich**: same log2 ratio for the loser/decoy gate-byte ($>$0=I2S corpus pinned to decoys)	signed_target_enrich $\geq$ 1.0
i2s_vp__LWL	*(inconclusive)*	*branches with no validated label*	35	—	—	—	—
i2s_vp__LW_	i2s_multibyte_literal_splice_vp_ gradient_blind	vpc's I2S enriches the exact multibyte literal in its corpus; value_profile's gradient does not	0	operand_enrichment	signed_target_enrich, decoy_ enrich	**signed_target_enrich**: log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive ($>$0=enriched, $<$0=depleted) **decoy_enrich**: same log2 ratio for the loser/decoy gate-byte ($>$0=I2S corpus pinned to decoys)	signed_target_enrich $>$ 0 AND decoy_enrich $<$ 0.5
i2s_vp__LW_	*(inconclusive)*	*branches with no validated label*	3	—	—	—	—
i2s_vp__WLW	decoy_overcommit_wrong_literal	vpc's I2S substitution pins the corpus to a WRONG decoy literal (sibling case / enumerated case / valid-range boundary), flooding the loser corpus with a value that blocks the target arm	0	operand_enrichment	decoy_enrich, signed_target_ enrich	**decoy_enrich**: same log2 ratio for the loser/decoy gate-byte ($>$0=I2S corpus pinned to decoys) **signed_target_enrich**: log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive ($>$0=enriched, $<$0=depleted)	decoy_enrich $>$ 0.8 AND signed_target_ enrich $<$ 0.3
i2s_vp__WLW	*(inconclusive)*	*branches with no validated label*	12	—	—	—	—
i2s_vp__WL_	vp_gradient_value_distance	value_profile's CMP_MAP gradient closes the value-distance to the case constants that vpc does not reach	0	value_distance_ reached	vp_min_distance, vpc_min_ distance, distance_gap	**vp_min_distance**: min Hamming/value distance from value_ profile seeds to the gate literal (0 = gate solved) **vpc_min_distance**: min Hamming/value distance from value_ profile_cmplog seeds to the gate literal (0 = gate solved) **distance_gap**: winner$-$loser gap in value/distance closed toward the gate	vp_min_distance $<$ vpc_min_distance AND distance_gap $\geq$ 2
i2s_vp__WL_	*(inconclusive)*	*branches with no validated label*	1	—	—	—	—

continued on next page

Shape	Cluster	Cluster description	#br	Tool(s)	Metric(s)	Metric meaning	Verify rule
i2s_vp__WWL	vp_cmpmap_gradient_distance_closure	VP's CMP_MAP distance-bucket feedback admits seeds that progressively CLOSE the byte-distance to the gate's comparison target (an exact multibyte literal, a grammar-token sequence, or a multi-field format constraint), so the VP-resolving corpus sits strictly closer to that target than the naive-blocking corpus – a continuous gradient naive's edge-only feedback cannot sense.	**5**	value_distance_reached	vp_min_distance, naive_min_distance, distance_closure_ratio	**vp_min_distance**: min Hamming/value distance from value_profile seeds to the gate literal (0 = gate solved) **naive_min_distance**: min Hamming/value distance from naive seeds to the gate literal (0 = gate solved) **distance_closure_ratio**: fraction of the value gap the winner closes (vs the loser/naive residual)	naive_min_distance $>$ 0 AND distance_closure_ratio $\geq$ 0.5
i2s_vp__WWL	vp_cmpmap_gradient_depth_reach	For gates with NO single fixed operand (digit-range loops, derived-integer arithmetic, deep multi-token parse reductions, multi-field format integrity), VP's gradient is visible as PARSE/STRUCTURE DEPTH: the VP-resolving corpus reaches a deeper parse state / longer in-range run / more-complete valid structure than the naive-blocking corpus, which stalls shallow.	0	depth_reach	vp_depth, naive_depth, depth_lift	**vp_depth**: max loop/recursion/path depth reached at the gate by the value_profile arm **naive_depth**: max loop/recursion/path depth reached at the gate by the naive arm **depth_lift**: max loop/recursion/path depth reached at the gate by the arm	depth_lift $\geq$ 1.5 AND naive_depth $<$ gate_depth_threshold
i2s_vp__WWL	*(inconclusive)*	*branches with no validated label*	9	—	—	—	—
i2s_vp__W_L	vp_basin_escape	VP CMP_MAP steers AWAY from a dominant magic-value basin: there is no single target literal — the true side wants ANY value but the corpus-dominant magic. VP-resolving seeds DEPART the basin (multi-modal: spread over several non-magic format magics), while naive-blocking seeds are pinned at the basin value.	**3**	value_distance_reached	naive_basin_lock_fraction, vp_basin_lock_fraction, vp_value_modality, basin_departure_lift	**naive_basin_lock_fraction**: naive fraction of seeds locked in the wrong value basin (gradient trap) **vp_basin_lock_fraction**: value_profile fraction of seeds locked in the wrong value basin (gradient trap) **vp_value_modality**: # distinct value modes value_profile explores at the gate **basin_departure_lift**: fraction of seeds locked in the wrong value basin (gradient trap)	basin_departure_lift $>$ 0 AND naive_basin_lock_fraction $\geq$ 0.7 AND vp_value_modality $\geq$ 3
i2s_vp__W_L	vp_structural_assembly_gradient	VP CMP_MAP closes DERIVED integer comparisons (table offsets/lengths computed from structural fields), not a fixed input literal; VP-resolving seeds carry a larger/more-valid parsed structure built from padding, with no fixed-operand distance signal.	0	struct_size_ratio*	vp_struct_size, naive_struct_size, struct_size_lift, vp_residual_distance, naive_residual_distance	**vp_struct_size**: value_profile median structural sub-record size at the gate **naive_struct_size**: naive median structural sub-record size at the gate **struct_size_lift**: winner/loser structural sub-record size ratio **vp_residual_distance**: value_profile remaining distance to the gate value after best approach **naive_residual_distance**: naive remaining distance to the gate value after best approach	struct_size_lift $\geq$ 1.3 AND distance_close_lift $<$ 0.25 * naive_residual_distance
i2s_vp__W_L	vp_direct_value_gradient	VP CMP_MAP climbs a monotone distance gradient toward a FIXED target value (single literal / switch constant / masked-equality / terminal byte of a chain); VP-resolving seeds reach a smaller residual distance to that target than naive-blocking seeds, which stall at a fixed distance.	0	value_distance_reached	vp_residual_distance, naive_residual_distance, distance_close_lift	**vp_residual_distance**: value_profile remaining distance to the gate value after best approach **naive_residual_distance**: naive remaining distance to the gate value after best approach **distance_close_lift**: winner $-$ naive distance closed toward the gate	distance_close_lift $>$ 0 AND vp_residual_distance $<$ naive_residual_distance (after sense-inversion for mismatch gates) AND abs(distance_close_lift) $\geq$ 0.25 * naive_residual_distance
i2s_vp__W_L	*(inconclusive)*	*branches with no validated label*	8	—	—	—	—
mopt_mutation_LW	mopt_scheduling_keyword_assembly_deficit	MOpt operator scheduling under-assembles the gate keyword that naive's havoc reaches	0	joint_necessity, token_count	mopt_tokens, naive_tokens, token_lift	**mopt_tokens**: # grammar keyword tokens placed per seed by the mopt arm **naive_tokens**: # grammar keyword tokens placed per seed by the naive arm **token_lift**: winner_tags $-$ loser_tags (structural-token advantage)	token_lift $\geq$ 1.5 AND mopt_tokens $<$ naive_tokens
mopt_mutation_LW	*(inconclusive)*	*branches with no validated label*	1	—	—	—	—
ngram_coverage_LW	ngram_shallow_corpus_saturation_starves_deep_path `[df]`	ngram-4 feedback floods the corpus with near-duplicate shallow-tokenizer seeds, starving energy from the rare deep xmlParseReference-$>$xmlAddChild path that naive reaches	0	corpus_size_ratio	corpus_size_ratio, depth_reach_inv	**corpus_size_ratio**: saved-corpus size (or I2S-arm/baseline size ratio) — corpus bloat **depth_reach_inv**: true if the deep path is NOT reached (inverted depth test)	corpus_size_ratio $\geq$ 1.5 AND depth_reach_inv == true
ngram_coverage_LW	*(inconclusive)*	*branches with no validated label*	1	—	—	—	—
ngram_coverage_WL	ngram_sequential_depth_reach	n-gram path-tuple retention lets naive_ngram4 reach deeper sequential-decode path states that plain edge coverage collapses and discards	**2**	depth_reach	winner_depth, loser_depth, depth_lift	**winner_depth**: max loop/recursion/path depth reached at the gate by the winner arm **loser_depth**: max loop/recursion/path depth reached at the gate by the loser arm **depth_lift**: max loop/recursion/path depth reached at the gate by the arm	depth_lift $\geq$ 1.5 AND winner_depth $>$ loser_depth
ngram_coverage_WL	*(inconclusive)*	*branches with no validated label*	2	—	—	—	—
TOTAL	**66 clusters** (27 non-empty / 25 distinct mech)	**258** val $+$ **363** inconcl $=$ **621** rows (554 distinct br)	**258**	**6 built + 3 unbuilt**	**built:** corpus_size_ratio, depth_reach, joint_necessity, operand_enrichment, token_count, value_distance_reached **unbuilt:** energy_share, struct_size_ratio, value_enrichment		